

I am a plant evolutionary geneticist whose goal is to understand the factors that facilitate and limit adaptation to environmental change. This objective is fundamental not just to the field of evolutionary biology, but to our ability to conserve endangered species and manage the spread of antibiotic, antiviral, and pesticide resistance. **My research thus aims to understand the evolutionary histories of adaptation, characterize the contributions and origins of complex genomic architectures, and assess the rate of change of adaptive genetic variation through space and time concomitant to contemporary landscape conversion.** To do so, I use cutting-edge population genomic approaches bolstered by temporal and spatial sampling designs, theoretical inference, and quantitative field and greenhouse experiments in non-model organisms.

The evolutionary histories of adaptation to extreme selection

The rapidly growing incidence of herbicide resistance across weed species—an impressive, if unplanned, evolutionary experiment costing billions per year—presents a pressing opportunity to understand factors that limit rapid adaptation so that we may manage resistance evolution and optimize food production in our changing climate. My work leverages this powerful natural experiment to characterize the full evolutionary history of adaptive alleles, considering environmental, organismal, and genomic sources of heterogeneity.

For most adaptations, the extent to which evolution occurs in parallel, the timescale over which causal mutations arise (de novo or from standing variation), and the role of gene flow in their spread, remain poorly known. I have developed *Amaranthus tuberculatus* as a tractable population genomic system to address these fundamental evolutionary questions in the context of herbicide resistance. *Amaranthus* is among the most problematic agricultural weeds in North America due to widespread resistance and crop yield reductions of up to 74% in soy and corn. In collaboration with international weed scientists and molecular geneticists, I generated the first high-quality genomic and population genomic resources for the species. I used these resources to reveal the interplay between demography and the origins of a complex copy number mutation conferring resistance to Roundup (glyphosate) herbicides (Kreiner et al. 2019, *PNAS*). I found that novel agricultural populations arose through both long-distance dispersal and introgression into local natural populations with multiple de novo mutations and gene flow underlying the spread of resistance.

Causal single-nucleotide mutations have been well characterized for herbicide resistance, but their evolutionary histories and relative importance compared to other genetic architectures had remained untested. By implementing cutting-edge coalescent inference approaches in *A. tuberculatus*, I found that large-effect resistance alleles repeatedly arose after the onset of herbicide use and showed evidence of dramatically strong selection across both recent and deep timescales (Figure 1) (Kreiner et al. 2022, *eLife*). Hyper-parallelism in the evolution of resistance was facilitated through a four-fold population size expansion over the last century that quelled mutation-limitation at single sites.

Yet even across populations with high frequencies of known large-effect resistance mutations, I found that the combined effects of many small-effect mutations across the genome (polygenicity) are of near-equal importance (Kreiner et al. 2021, *Molecular Ecology*). More than 250 genes across all 16 chromosomes contribute to resistance, show signals of recent selection, and appear constrained by pleiotropic effects on reproduction and growth. My results paint a picture of the rise, spread, and fate of adaptive alleles of varying complexity in the face of

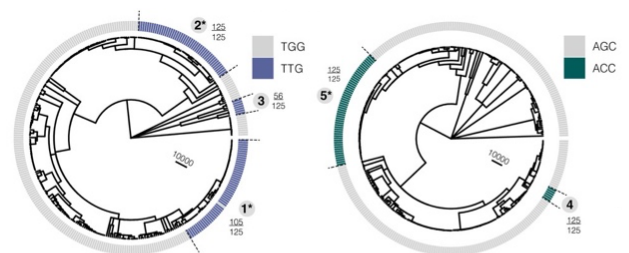


Figure 1. Haplotype trees illustrating multiple origins and strong selection on two common mutations conferring resistance to ALS herbicides.

extreme selection. *A. tuberculatus* appears to epitomize the qualities of a species that persists despite all odds: one where the imagined dichotomies of adaptation from small vs. large effect alleles, recurrent origins vs. gene flow, old vs. young mutations are revealed as false.

Future work on resistance evolution in my lab will explore the following questions:

What factors control the genetic architecture of adaptation? Central to the goal of predicting and managing dynamic adaptive processes is the degree to which the genetic architectures of adaptation vary across populations and species. A master's-level project will investigate the extent to which alleles conferring resistance do so consistently by quantifying the architecture of resistance independently across several populations. A Ph.D. or postdoctoral project will combine spatial genomics and modelling of monogenic and polygenic architectures to test how landscape patchiness of herbicide applications (selection) and crop field proximity (gene flow) influence the types of genetic changes (large vs. small effect mutations) that occur and persist.

How does life history shape the rate and nature of adaptation? Through a phylogenetic meta-analysis, I found that populations of selfing species evolve to herbicides in a more mutation-limited framework than outcrossing species, an effect that may be alleviated by polyploidy (Kreiner et al. 2018, *Annual Review of Plant Biology*). To understand the mechanisms underlying adaptive constraints across diverse plant lineages, a long-term aim will be to clarify how life-history traits influence the dynamics of rapid adaptation through comparative population genomics and complementary evolve and resequence experiments that incorporate selfing rate manipulations and induction of synthetic polyploids via colchicine.

The timescales of adaptation to anthropogenic landscapes

In addition to continuing research on adaptation at selective extremes, my group will study the evolutionary response of plants to the continuing conversion of our landscape. Such extensive environmental change offers the chance to understand the interplay between demographic and adaptive processes across scales—from the timing and spatial origins of expansions, contractions, and adaptive shifts, to the strengths of selection underlying change in genetic variation for fitness.

My ongoing research dissects these signatures of contemporary adaptation to agriculture, from genotype to phenotype, through space and time, leveraging the unique *Amaranthus tuberculatus* system in which the secondary contact of two native subspecies shapes diversity across the range.

Across this range, I performed extensive field sampling, common garden phenotyping, and whole-genome sequencing of 34 paired natural and agricultural populations along with >100 herbarium specimens spanning the last 200 years of agricultural-land use change (Figure 2).

I performed a manipulative common garden experiment on 6000 individuals from these paired-environmental collections, and found a strong association between ancestry, flowering time, and

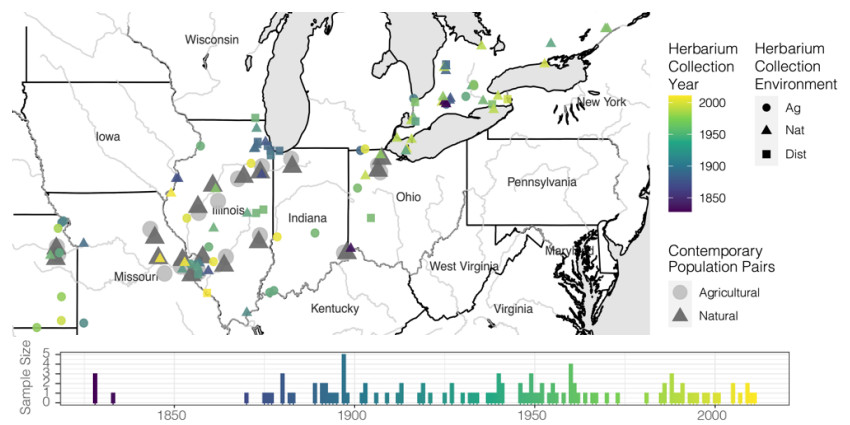


Figure 2. Map and temporal distribution of sequenced herbarium samples spanning environment types along with contemporary collections of 34 paired natural and agricultural populations.

biomass across environments, highlighting that the *A. tuberculatus* agricultural invasion is facilitated by subspecies-specific standing genetic variation (Kreiner et al., 2021, *Evolution*). To quantify the strength and tempo of these evolutionary processes, I developed a novel method for estimating selection coefficients and allele-frequency trajectories on the time series data like those contained within our herbarium specimens. I found that while the frequency of modern agricultural alleles has increased considerably over the last two centuries, the intensification of agriculture over the last six decades has driven strengths of selection much stronger than typically observed ($s=0.054$), with effects pervasive even in natural habitats ($s=0.026$) (Figure 3) (Kreiner et al., second revisions in review at *Science*). Over this same timescale, I directly observe shifts in ancestry through time—a human-mediated range expansion that provided the standing genetic variation on which agricultural selection could act. This work highlights the novel application of sequencing herbarium collections to directly quantify temporal adaptive dynamics, and demonstrates how anthropogenic change not only reshapes environments but provides an opportunity for range expansion to facilitate and feedback with local adaptation.

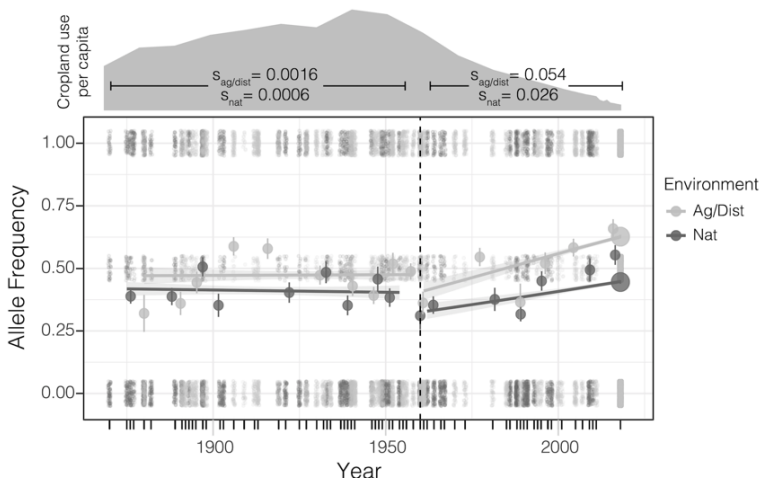


Figure 3. Environment-specific selection estimates and joint allele frequency trajectories for 154 modern agricultural alleles before and after the 1960's onset of agricultural intensification (reflected in the decrease in cropland use per capita) across 108 herbarium samples.

Future work in the lab will expand on these approaches for characterizing the rate and timescale of adaptation to contemporary environments. First, we will perform long-read population resequencing on our paired collections to identify the role of structural variants in agricultural adaptation and identify haplotype blocks as markers to track through time within our herbarium sequences. By comparing estimates of selection on these structural variants to SNPs (as in Figure 3), we will address the outstanding question of how the distribution of selection coefficients for adaptation varies with mutational complexity. Next, a Ph.D. student will perform multi-year field experiments to uncover how rapidly

chemical and climate-based selective pressures such as herbicides and drought drive change in phenotypic and genomic variation within and across seasons. By comparing to decade- and century-long historical inferences of the rate and direction of evolution, this work will address the longstanding enigma of the consistency of evolutionary change across timescales.

Vision for the Kreiner Lab

My work strives to understand how plants adapt and persist in the face of anthropogenic change while addressing key outstanding questions in evolutionary genetics. By bridging a gap between population genomics and weed science, the projects I have outlined above will also contribute to the applied goals of improved management of agricultural weeds and efficient food production in a changing climate. Spanning fundamental and applied sciences, my integrative research program not only lends well to collaboration with researchers from diverse backgrounds, including molecular, ecological, agricultural, and genomic, but strengthens prospects for funding from USDA, DOE, NSF, and NIH. My research group will foster an inclusive, collaborative, and engaging community, which I expect to flourish in the outstanding atmosphere at University of Chicago.